Recap

Resolvers are software components that implement the DNS protocol

Full resolvers run on every Name Server of the DNS hierarchy and can interact with any other resolver

Stub resolvers run on every user device and can only interact with the full resolvers of their Local Name Server using recursive queries

Iterated queries put the entire responsibility on the Local Name Server

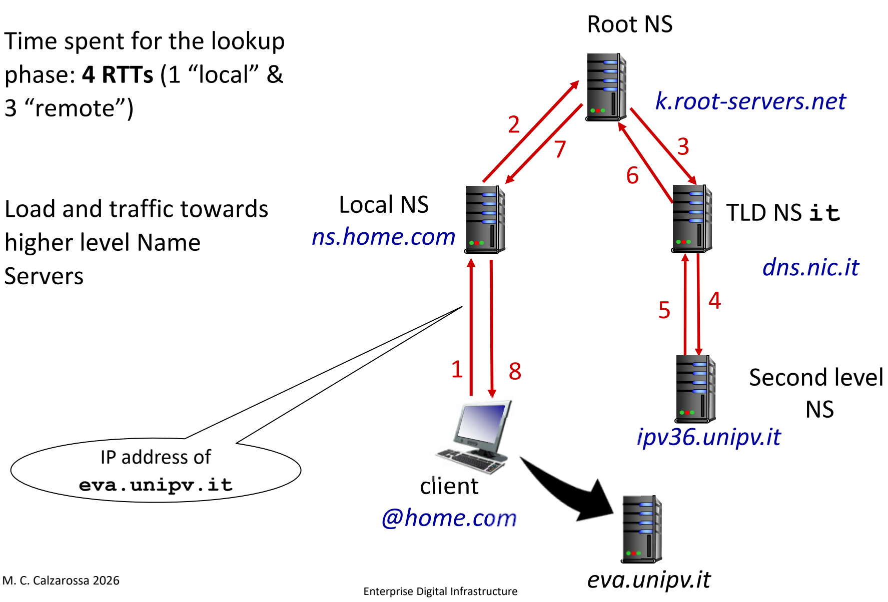
Queries involving Root Name Servers and TLD Name Servers are iterated

Authoritative responses are only those provided by the domain owners

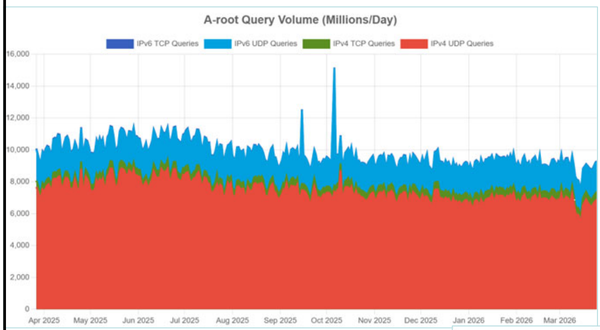
Example of recursive query

Time spent for the lookup phase: **4 RTTs** (1 “local” & 3 “remote”)

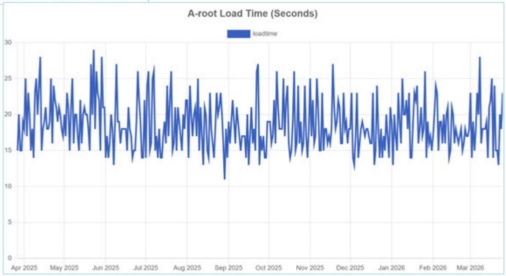
Load and traffic towards higher level Name Servers



Traffic at `a.root-servers.org`



[Source: https://a.root-servers.org/metrics]



Root Name Servers Performance

Location: World | Type: Raw Performance | Uptime | Quality | Period: Last 30 days

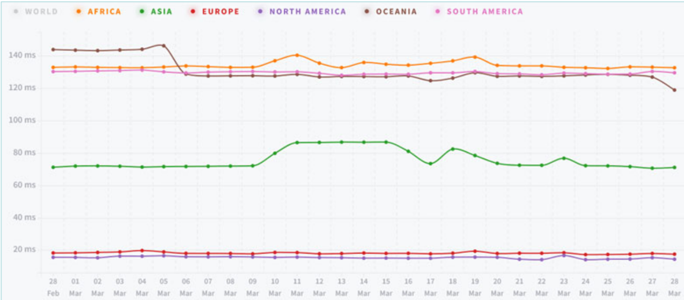
	DNS name	Query Speed
1	f.root-servers.net	7.19 ms
2	e.root-servers.net	13.43 ms
3	d.root-servers.net	21.64 ms
4	j.root-servers.net	23.23 ms
5	h.root-servers.net	40.02 ms
6	i.root-servers.net	45.25 ms
7	a.root-servers.net	46.17 ms
8	k.root-servers.net	50.07 ms
9	l.root-servers.net	51.82 ms
10	c.root-servers.net	52.59 ms
11	m.root-servers.net	63.26 ms
12	g.root-servers.net	67.43 ms
13	b.root-servers.net	71.33 ms

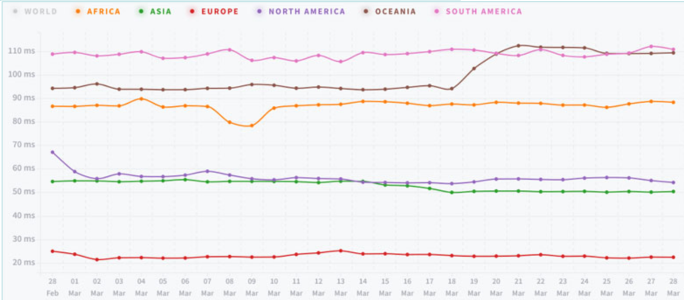
World | Uptime

	DNS name	Uptime
1	d.root-servers.net	99.94 %
2	l.root-servers.net	99.94 %
3	a.root-servers.net	99.91 %
4	c.root-servers.net	99.91 %
5	h.root-servers.net	99.9 %
6	g.root-servers.net	99.89 %
7	b.root-servers.net	99.87 %
8	e.root-servers.net	99.6 %
9	m.root-servers.net	99.48 %
10	i.root-servers.net	99.46 %
11	f.root-servers.net	99.03 %
12	k.root-servers.net	98.59 %
13	j.root-servers.net	97.86 %

[Source: https://www.dnsperf.com]

a & **k** Root servers





Enterprise Digital Infrastructure

[Source: https://www.dnsperf.com]

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Public DNS Performance

Location: Europe | Type: Raw Performance | Uptime | Quality | Period: Last 30 days

	DNS name	Query Speed
1	1.1.1.1	5.87 ms
2	FlashStart	8.92 ms
3	Quad9	10.33 ms
4	Google	10.89 ms
5	Control D	11.98 ms
6	Cisco Umbrella	12.81 ms
7	Gcore	13.39 ms
8	DNSFilter	13.97 ms
9	Lineguard by Secure64	15.9 ms
10	SafeDNS	16.53 ms
11	DNS4EU (EU only)	17.02 ms
12	NextDNS	17.54 ms
13	UltraDNS	21.11 ms

Europe | Uptime

	DNS name	Uptime
1	1.1.1.1	99.97 %
2	Google	99.97 %
3	Control D	99.96 %
4	FlashStart	99.96 %
5	Gcore	99.96 %
6	Lineguard by Secure64	99.96 %
7	NextDNS	99.96 %
8	DNSFilter	99.94 %
9	SafeDNS	99.92 %
10	DNS4EU (EU only)	99.89 %
11	Quad9	99.79 %
12	Cisco Umbrella	96.86 %
13	UltraDNS	95.07 %

[Source: https://www.dnsperf.com]

Enterprise Digital Infrastructure

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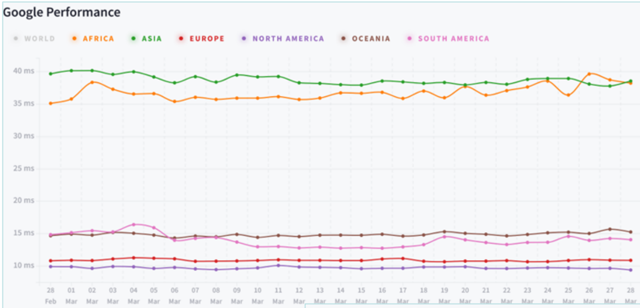
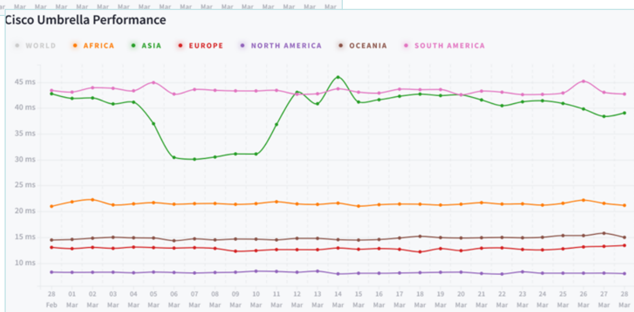
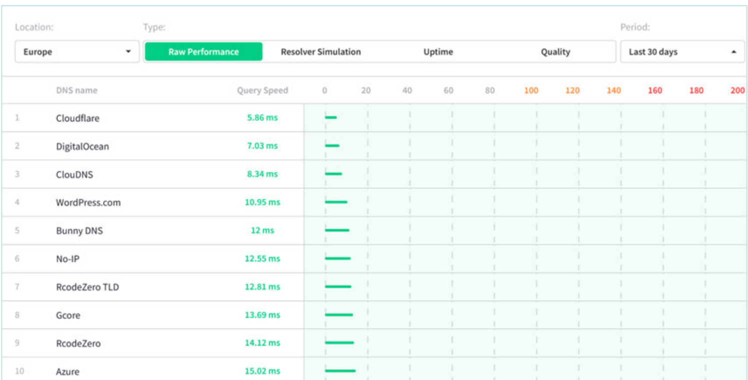
Google vs Cisco





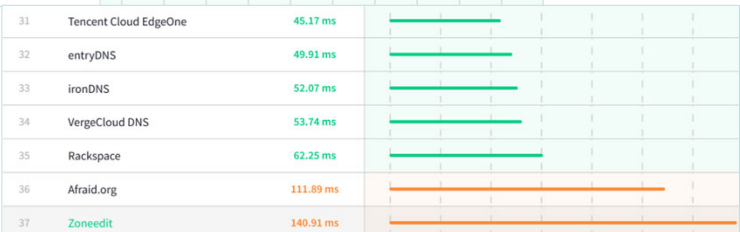
[Source: https://www.dnsperf.com]

Authoritative DNS Providers



Location: Europe | Type: Raw Performance | Resolver Simulation | Uptime | Quality | Period: Last 30 days

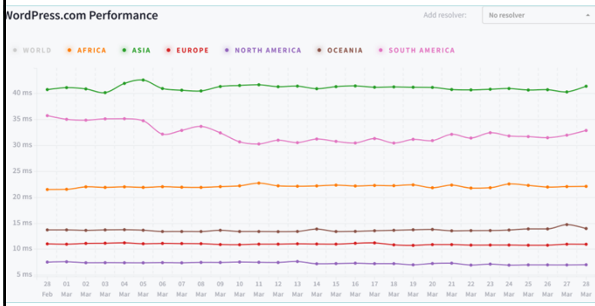
	DNS name	Query Speed
1	Cloudflare	5.86 ms
2	DigitalOcean	7.03 ms
3	ClouDNS	8.34 ms
4	WordPress.com	10.95 ms
5	Bunny DNS	12 ms
6	No-IP	12.55 ms
7	RcodeZero TLD	12.81 ms
8	Gcore	13.69 ms
9	RcodeZero	14.12 ms
10	Azure	15.02 ms

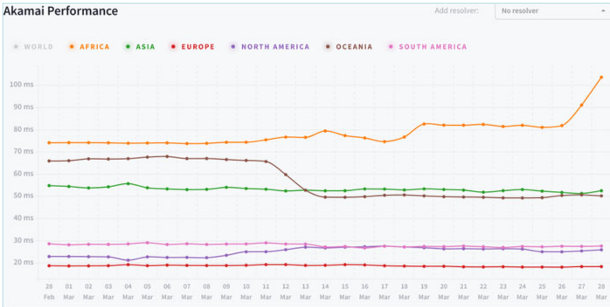


31	Tencent Cloud EdgeOne	45.17 ms
32	entryDNS	49.91 ms
33	ironDNS	52.07 ms
34	VergeCloud DNS	53.74 ms
35	Rackspace	62.25 ms
36	Afraid.org	111.89 ms
37	Zoneedit	140.91 ms

[Source: https://www.dnsperf.com]

WordPress vs Akamai





[Source: https://www.dnsperf.com]

DNS Messages

A protocol has to specify the *types* of messages exchanged in the interactions between clients and servers as well as the *syntax* of these messages

Two types of DNS messages

1. Query messages
2. Response messages

Both types of message share the same format

Messages are characterized by variable length

Information are subdivided into *five* sections

One section, i.e., header section, is common to both types of messages

DNS message format

Header section (12 bytes) includes:

1. ID used to match query and response
2. Flags
   - query or response
   - recursion desired
   - recursion available
   - authoritative answer
   - ….
3. Number of questions and number of RRs in the answer/authority/ additional sections

header section

2 bytes	2 bytes
identifier	flags
# questions	# answer RRs
# authority RRs	# additional RRs
question (variable # of questions)	
answer (variable # of RRs)	
authority (variable # of RRs)	
additional info (variable # of RRs)	

DNS message format (cont'd)

Question section: contains the query

qname, qclass, qtype

unipv.it IN NS

poldo.unipv.it IN A

2 bytes	2 bytes
identifier	flags
# questions	# answer RRs
# authority RRs	# additional RRs
question (variable # of questions)	
answer (variable # of RRs)	
authority (variable # of RRs)	
additional info (variable # of RRs)	

DNS message format (cont'd)

Answer section: contains RR(s) in response to the query

Authority section: contains RRs of the Authoritative Name Server of the domain

Additional info section: contains RRs that might be useful (even though not asked in the question)

2 bytes	2 bytes
identifier	flags
# questions	# answer RRs
# authority RRs	# additional RRs
questions (variable # of questions)	
answers (variable # of RRs)	
authority (variable # of RRs)	
additional info (variable # of RRs)	

Examples

```
;; QUESTION SECTION:
;ucsd.edu.                      IN      NS

;; ANSWER SECTION:
ucsd.edu.               86400   IN      NS      ns-auth1.ucsd.edu.
ucsd.edu.               86400   IN      NS      ns-auth2.ucsd.edu.
ucsd.edu.               86400   IN      NS      ns-auth3.ucsd.edu.

;; ADDITIONAL SECTION:
ns-auth1.ucsd.edu.      172800  IN      A       44.227.162.249
ns-auth2.ucsd.edu.      172800  IN      A       132.239.252.172
ns-auth3.ucsd.edu.      172800  IN      A       132.239.252.186

;; Query time: 190 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sun Mar 29 16:41:42 CEST 2026
;; MSG SIZE  rcvd: 154
```

```
;; QUESTION SECTION:
;harvard.edu.                   IN      NS

;; ANSWER SECTION:
harvard.edu.            244     IN      NS      a10-66.akam.net.
harvard.edu.            244     IN      NS      a16-64.akam.net.
harvard.edu.            244     IN      NS      a11-67.akam.net.
harvard.edu.            244     IN      NS      a6-66.akam.net.
harvard.edu.            244     IN      NS      a7-65.akam.net.
harvard.edu.            244     IN      NS      a1-171.akam.net.

;; ADDITIONAL SECTION:
a7-65.akam.net.         244     IN      AAAA    2600:1406:32::41
a11-67.akam.net.        244     IN      A       84.53.139.67
a7-65.akam.net.         244     IN      A       23.61.199.65
a6-66.akam.net.         244     IN      AAAA    2600:1401:1::42
a10-66.akam.net.        244     IN      A       96.7.50.66
a10-66.akam.net.        244     IN      AAAA    2600:1480:d000::42
a11-67.akam.net.        244     IN      AAAA    2600:1480:1::43
a6-66.akam.net.         244     IN      A       23.211.133.66

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sun Mar 29 16:48:24 CEST 2026
;; MSG SIZE  rcvd: 348
```